Technical Document #721: Blockchain - Comprehensive Overview

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Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Cross-chain technology enables communication between different blockchains. It is essential for interoperability in the blockchain ecosystem.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Smart contracts are self-executing contracts with terms directly written into code.
- Proof of Stake (PoS) selects validators based on their stake in the network.
- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.